SAP CAP Backend Developer

Practical Mock Exam 01 — CAP Procurement Management

Unofficial practice scenario • Not an actual or leaked SAP exam task • Suggested practice time: 180 minutes

Your assignment

A company needs a CAP backend to manage suppliers, products, and purchase orders. You have been assigned an existing or newly created CAP Node.js project. Implement the required functionality, run the tests, and prepare it for deployment.

Objective: Deliver a working application that meets every requirement below. Writing code without verifying its behavior is insufficient.

Task 1 — Data modeling (15 points)

Create the CDS data model for the following entities:

Entity	Required fields
Suppliers	ID, name, email, active status
Products	ID, name, description, price, stock quantity, supplier
PurchaseOrders	ID, order date, status, supplier
OrderItems	ID, quantity, unit price, product, purchase order

- Use appropriate CDS types and keys.
- Model the relationships between suppliers, products, orders, and order items.
- Make the order-to-items relationship suitable for a purchase order containing multiple items.
- Apply suitable mandatory-field and validation constraints.

Task 2 — Expose the service (10 points)

- Create a service named **ProcurementService** that exposes the required entities.
- The service is available through the local CAP server.
- The service document and entity sets can be retrieved.
- CRUD operations work for entities where appropriate.
- Relationships can be navigated through the service.

Task 3 — Implement business logic (20 points)

- Reject a product with a negative price.
- Reject an order item with a quantity less than one.
- Reject a purchase order if its supplier is inactive.
- Calculate the order total from its items.
- Prevent an order from being submitted if it contains no items.
- Implement relevant CAP event handlers and verify that invalid requests fail correctly.

Task 4 — Add a custom operation (10 points)

- Implement a business operation named **submitOrder**.
- Accept a purchase order ID.

• Verify that the order exists and is eligible for submission.

• Update its status to **SUBMITTED**.

• Return a meaningful result or error.

• Prevent invalid status transitions.

• Choose an appropriate CDS action or function and implement it using CAP conventions.

Task 5 — Authorization (10 points)

• Define an appropriate authorization role for procurement managers.

• Restrict order submission to authorized users.

• Ensure protected operations cannot be accessed merely by hiding buttons in the UI.

• Test both authorized and unauthorized requests.

• Use authentication and authorization configuration appropriate to your CAP runtime environment.

Task 6 — Persistence and query behavior (10 points)

• Verify that records persist using your configured database.

• Filter products by supplier.

• Sort products by price.

• Retrieve an order together with its related items.

• Ensure invalid data does not leave inconsistent order records.

Task 7 — Build and deployment (15 points)

• Prepare the application for SAP BTP Cloud Foundry deployment.

• Configure the required MTA modules and resources.

• Run the applicable build command and generate the deployment archive.

• Deploy using the appropriate Cloud Foundry workflow, where your account and environment permit it.

• Inspect application logs if deployment fails.

• Local success does not count as proof of successful cloud deployment.

Task 8 — Final verification (10 points)

• The CAP server starts without errors.

• Service metadata and entity endpoints respond.

• Valid data can be created and retrieved.

• Invalid input is rejected.

• The custom order operation behaves correctly.

• Authorization restrictions work.

• Build and deployment status are documented.

Scoring rubric

Area	Points
CDS data modeling	15
Service exposure	10
Business logic	20
Custom operation	10
Authorization	10
Persistence and queries	10
Build and deployment	15
Verification	10
TOTAL	100

This scoring rubric is created for practice; it is not SAP's official scoring scheme or passing threshold.

How to use this mock exam

Work in VS Code or SAP Business Application Studio. Implement Task 1 first, run the relevant checks, then continue task by task. Do not look for a ready-made solution before attempting the work. For official preparation, verify the current exam format and scope on SAP's certification page.

Official references: https://learning.sap.com/certifications/sap-certified-associate-backend-developer-sap-cloud-programming-model and https://learning.sap.com/helpcenter/certification-support/system-based-assessments-faq